

Hyperspectral coded events

A diffractive-surface model for spectrum recovery, with corrected information bounds

Goal. Combine the wavelength-dependent, spatially varying diffractive encoding of Majumder et al. [1] with event-based measurement and information-driven optical design as in CodedEvents [2]. The unknown is a *spectrum or spectral image cube*, not just one wavelength. The equations below are a proposed synthesis; the cited papers do not establish this combined system.

1. Diffractive encoding of a spectral scene

For mutually incoherent scene points and wavelength bands, let $s_{rm} \geq 0$ be the band coefficient at spatial cell r and spectral band m . Stack these into $\mathbf{s} \in \mathbb{R}^D$, $D = RK$. Over a short estimation window, use a fixed reference cube and a specified temporal mechanism (known motion, modulation, or a finite-dimensional dynamics model):

$$I_j(t) = B_j(t) + \sum_{r=1}^R \sum_{m=1}^K H_{j,rm}(t; \alpha, \nu) s_{rm} = B_j(t) + \mathbf{h}_j(t)^T \mathbf{s}. \quad (1)$$

Here I_j is mean photocurrent or photoelectron rate, B_j is additive background, ν collects uncertain motion/calibration parameters, and α is the *known design* of the surface and sensor gap. One calibrated scalar-diffraction construction is

$$H_{j,rm}(t) = \int \psi_m(\lambda) \eta_j(\lambda) \int_{\mathcal{P}_j} |\mathcal{P}_{\lambda,z}\{E_{r,\lambda,t} T_\alpha\}(\mathbf{u})|^2 d\mathbf{u} d\lambda, \quad (2)$$

$$T_\alpha(\boldsymbol{\rho}, \lambda) = a_\alpha(\boldsymbol{\rho}, \lambda) \exp\left[\frac{2\pi i}{\lambda} \Delta n(\lambda) d_\alpha(\boldsymbol{\rho})\right].$$

ψ_m is a normalized band basis; $\mathcal{P}_{\lambda,z}$ propagates the field to the sensor; $\mathcal{P}_j$ is pixel area. Normalize the input field per unit coefficient. Retain throughput, wavelength scaling and space variance; a near-sensor DFA is not generally a shift-invariant far-field PSF. A calibrated vector model can replace scalar propagation for a metasurface or high-NA system. **Sum optical intensities before taking the logarithm.**

2. What makes the spectrum visible to events?

The ideal contrast signal is $L_j = \log I_j$. An event occurs at the first crossing of $L_j - R_j = +C_{j,+}$ or $-C_{j,-}$, then the pixel reference R_j updates. Thresholds, reset and filtering matter; events are not individual photons.

A static spectrum and static optics produce no intended contrast events. Moreover, if only a common brightness factor changes and $B_j = 0$,

$$I_j(t) = a(t)(H\mathbf{s})_j \implies q_j \equiv \partial_t \log I_j = \dot{a}/a, \quad \nabla_s q_j = \mathbf{0}. \quad (3)$$

Thus common blinking alone does not preserve the static spatial code in ideal log contrast when the reference scales with that signal. This does not cover a calibrated onset from a nonzero reference, or separately modeled intensity-dependent noise.

Two usable mechanisms: (i) gate the scene relative to a calibrated, nonzero additive reference; or (ii) move/dither the scene or coded pattern so that $H(t)$ changes non-proportionally. The latter can retain spectral *shape*, but pure contrast with zero background still cannot determine global brightness. Both mechanisms are developed below.

Event likelihood and spectral Fisher information

3. The spectral sensitivity that must be optimized

Hold the reference-cube coefficients $\mathbf{s}$ constant during the modeled temporal probe. Differentiating (1) at fixed nuisance parameters gives

$$\dot{I}_j = \dot{B}_j + \dot{\mathbf{h}}_j^T \mathbf{s}, \quad \mathbf{u}_j \equiv \nabla_{\mathbf{s}} q_j = \frac{I_j \dot{\mathbf{h}}_j - \dot{I}_j \mathbf{h}_j}{I_j^2}. \quad (4)$$

This quotient-rule term is the spectral analogue of a localization derivative. It measures changes in the *normalized temporal response*, not simply differences between monochromatic intensity PSFs. Known motion is encoded in $\dot{H}$; for a translated intensity PSF its contribution is $-\mathbf{v} \cdot \nabla I$.

4. Correct likelihood for timestamped, signed events

For events $e_k = (j_k, p_k, t_k)$ on a fixed interval $[0, T]$, define the conditional event rate $r_{jp}(t \mid \mathcal{H}_t; \theta)$, where $p \in \{-1, +1\}$, $\theta = (\mathbf{s}, \nu)$ and $\mathcal{H}_t$ is the history before t . Fix or marginalize the initial reference state. A regular simple point process has

$$\ell(\theta) = \sum_k \log r_{j_k p_k}(t_k \mid \mathcal{H}_{t_k}; \theta) - \sum_{j,p} \int_0^T r_{jp}(t \mid \mathcal{H}_t; \theta) dt, \quad (5)$$

$$U \equiv \nabla_{\theta} \ell = \sum_{j,p} \int_0^T \nabla_{\theta} \log r_{jp} [dN_{jp} - r_{jp} dt]. \quad (6)$$

Compensated increments have zero conditional mean and variance $r_{jp} dt$; distinct marks have zero quadratic covariation. Consequently,

$$\mathcal{J}(\theta) = \mathbb{E}[UU^T] = \mathbb{E} \left[\sum_{j,p} \int_0^T \frac{\nabla_{\theta} r_{jp} \nabla_{\theta} r_{jp}^T}{r_{jp}} dt \right]. \quad (7)$$

The survival integral in (5) includes information from *no events*. Equations (5)–(7) require a differentiable correct likelihood, common support, finite information and valid expectation/derivative interchange. They are not a hardware noise model by themselves.

5. A computable approximation, explicitly labeled

For design exploration, approximate the marked streams as independent inhomogeneous Poisson processes with a calibrated smooth response

$$r_{jp}(t) = \rho_{jp}(I_j, q_j), \quad \mathbf{v}_{jp} \equiv \nabla_{\mathbf{s}} r_{jp} = \rho_{jp, I} \mathbf{h}_j + \rho_{jp, q} \mathbf{u}_j, \quad \mathcal{J}_{ss} = \sum_{j,p} \int_0^T \frac{\mathbf{v}_{jp} \mathbf{v}_{jp}^T}{r_{jp}} dt. \quad (8)$$

A coarse choice is $\rho_{jp}(I, q) = b_{jp}(I) + g(pq)/C_{jp}$ with calibrated nonnegative smooth g . The limit $g(x) = [x]_+$ approximates monotonic changes spanning many thresholds, only away from zero slope; it fails for subthreshold oscillations and refractory dynamics. Retain $b'(I)$ if noise depends on illumination. If the model also depends on threshold, bandwidth or motion parameters, differentiate those and include their nuisance blocks. Validate this approximation against a noisy threshold simulator or calibrated event data [3].

Recovering the spectrum and designing the surface

6. A concrete reference-based acquisition

For a first controlled experiment, keep H fixed, gate the scene with known $a(t)$, and keep a calibrated reference $B_j > 0$ *outside the gate*. With $f_j = (Hs)_j$,

$$I_j = B_j + af_j, \quad q_j = \frac{\dot{a}f_j}{B_j + af_j}, \quad \nabla_s q_j = \frac{\dot{a}B_j}{(B_j + af_j)^2} \mathbf{h}_j. \quad (9)$$

This restores spectral sensitivity. The reference is a proposed acquisition choice, not a claim that added background always helps. Its shot noise and the pixel bandwidth must be included in calibration.

For an ideal settled transition from $a = 0$ to $a = 1$, signed events approximate

$$y_j \equiv \sum_{k \text{ at } j} p_k C_{j,p_k} \simeq \log \frac{B_j + f_j}{B_j}, \quad \tilde{f}_j = B_j(e^{y_j} - 1). \quad (10)$$

The discrepancy is the difference of the terminal and initial contrast residuals. For symmetric threshold C its magnitude is below $2C$ in the ideal model, or below C if the reference is aligned at the start. Real noise, missed events and latency add error. Never set the baseline to zero inside the logarithm. Recovering $\tilde{\mathbf{f}}$ and applying the DFA inverse problem is a useful initialization; the nonlinear transform makes its noise nonuniform.

7. Decoder, observability and the correct bound

A statistically consistent target is the calibrated event likelihood, with a stated prior when needed:

$$(\hat{\mathbf{s}}, \hat{\nu}) = \arg \min_{\mathbf{s} \geq 0, \nu} \{-\ell(\mathbf{s}, \nu; \alpha) + \beta \mathcal{R}(\mathbf{s})\}. \quad (11)$$

Use spatial/spectral TV, a low-rank model or a spectral basis only as explicit structural assumptions. Under a fixed H and scalar gating, all data depend on $\mathbf{s}$ through $H\mathbf{s}$: more events cannot remove $\ker H$. A full cube therefore needs adequate independent temporal coding or a justified lower-dimensional scene model. Check the rank of the *event* Jacobian, not merely PSF dissimilarity.

With unknown nuisance parameters, the effective local information and unbiased-estimator bound are

$$\mathcal{J}_{\text{eff}} = \mathcal{J}_{ss} - \mathcal{J}_{sv} \mathcal{J}_{vv}^{-1} \mathcal{J}_{vs}, \quad \text{Cov}(\hat{\mathbf{s}}) \geq \mathcal{J}_{\text{eff}}^{-1}. \quad (12)$$

Invert only identifiable, appropriately scaled coordinates. With zero reference and brightness-independent event noise, $\mathbf{s} \mapsto c\mathbf{s}$ leaves ideal contrast unchanged even under dither: estimate normalized spectral shape, fix a scale gauge, or add an intensity anchor. Apply (12) in independent coordinates after any constraints. Regularization biases an estimator; its Hessian is not additional classical measurement Fisher information.

8. Optical objective and the practical test

Optimize the surface, gap and temporal probe for spectral information, e.g.

$$\min_{\alpha, \text{probe}} \mathbb{E}_{\text{design scenes}} [\text{tr}\{W \mathcal{J}_{\text{eff}}^{-1}\}], \quad (13)$$

with specified spectral weights $W \geq 0$, a fixed photon budget, fabrication limits and event-rate/bandwidth constraints. Use only identifiable target coordinates. First calibrate narrowband spatial responses and event dynamics; then test held-out spectral mixtures against a spectrometer. Comparing reference gating and dither will show whether the designed spatial code survives the event conversion.

Concise audit of the original CRLB.pdf

Equation numbers below refer to Rich Baird’s supplied 10-page manuscript. The optical/spectral model on pages 1–3 replaces its single-wavelength/localization formulation.

Original location	Correction and consequence
Eq. (14)	OFF events require $\Delta \log I \leq -C$, not $\geq -C$. Define the preceding reference and reset at each pixel.
Eq. (16)	Separate fixed-pixel and material derivatives. Translation contributes $-\mathbf{v} \cdot \nabla I$; intrinsic brightness or spectral changes require their own terms.
Eqs. (17)–(22)	Poisson count algebra is correct, but photon arrivals do not establish a Poisson event likelihood. A noiseless log ramp produces periodic events. If C is contrast threshold, coarse event frequency scales as $1/C$.
Eqs. (23)–(26)	Include the sign derivative and any intensity dependence of noise/sensitivity. For fixed background and sensitivity the sign cancels in FIM products away from zero. Use timestamps, polarity and a time integral for full-stream information.
Eqs. (15), (27)	A spectrum requires multiple spectral coefficients and an incoherent sum before the log. A DFA can be space variant and near-field. A nine-entry symbolic matrix is not a derived hyperspectral system matrix.
Eqs. (27)–(28); Fig. S1	Do not treat a derived intensity gradient as a free scalar, or known mask design as an estimated unknown. Test identifiability. The reported condition number 3.9×10^{19} requires rescaling and rank analysis before inversion. FIM and inverse-FIM correlations differ.
Eqs. (29)–(30); Fig. S2	Brightness cancellation is conditional and is not a noise law. No universal $\text{CRLB} \propto N$ or low-light optimum follows. Source intensity, additive background, threshold noise and bandwidth must be separated.
Page 6; CodedEvents	Longer observation with retained events cannot reduce information about fixed parameters. Binning can discard timing information. CodedEvents models two-time positions and examines speed; it is not simply a Gaussian intensity-difference model [2].
Eq. (12); introduction	For unbiased estimates, $0 = \nabla_{\theta} \mathbb{E}[\hat{\theta} - \theta] = \mathbb{E}[(\hat{\theta} - \theta)U^T] - I$. Bits/nats specify log base; Fisher information is local KL curvature, not quantized Shannon information.

References and scope

- [1] A. Majumder et al., “High-definition (HD) snapshot diffractive computational spectral imaging and inferencing,” *Optica* **12**, 1539–1547 (2025). doi:10.1364/OPTICA.559279. Forward-operator and calibration details consulted in the authors’ open preprint, Working Principle and Methods; the publisher full text was unavailable to the reader used here.
- [2] S. Shah et al., “CodedEvents: Optimal Point-Spread-Function Engineering for 3D-Tracking with Event Cameras,” CVPR (2024), Sec. 3.2 and Supplement S3. arxiv.org/abs/2406.09409.
- [3] R. Graça, B. McReynolds and T. Delbruck, “Shining Light on the DVS Pixel: A Tutorial and Discussion About Biasing and Optimization,” CVPR Workshops (2023). arxiv.org/abs/2304.04706.

Verification: Eq. (4) agreed with central differences to 1.8×10^{-10} relative error in a three-band test; zero-reference scale invariance and the gating limit were also checked. No device-specific spectral resolution or numerical CRLB is claimed without a calibrated optical operator and event likelihood. All equations are vector typeset.